

GeoExplain architecture

Package geoxplain: modular layers from data to explained, georeferenced outputs

Interfaces

Python API

CLI
geoxplain ...

Workflows
run_experiment

QGIS Processing
(optional)

Explainability

explainers
SHAP · permutation
PDP · local

spatial
spatial SHAP
prediction · uncertainty

validation
random · spatial split
block CV · Moran's I

metrics
classification
regression

Core

models
RF · XGBoost · LightGBM
MLP · LogReg · SVM

io
CSV · GeoTIFF
CRS-preserving

datasets
synthetic scenes

utils
reproducibility · errors

Data in / results out

Point tables
CSV: x, y,
features, target

Raster stacks
GeoTIFF,
shared grid

Metric tables
CSV · Excel

Prediction,
uncertainty,
SHAP GeoTIFFs

Figures
PNG 300 dpi
SVG · PDF

HTML report
reproducibility
record